



React JS Full Stack Development with PostgreSQL

(React JS + Express JS + PostgreSQL + JWT + Redux Toolkit + Razorpay)



Course Duration

1.5 Hrs to 2 Hrs / Session

5 Days / Week

30-35 Days / 45-50 Hours (Flexible)



Course Objective

This industry-oriented hands-on Training Program is designed to help students build modern, production-ready web applications using React JS and the PERN Stack. The course starts with JavaScript ES6 fundamentals and gradually progresses to authentication, REST APIs, state management, payment integration, and a complete E-Commerce application.

By the end of this course, participants will be able to design, develop, and deploy full-stack React applications using industry best practices.



Course Modules

Module 1 – JavaScript ES6+ Fundamentals

Build a strong foundation in modern JavaScript.

Topics Covered

- Variables (`let` & `const`)
- Arrow Functions
- Template Literals
- Destructuring
- Spread & Rest Operators
- ES Modules
- Promises

- Async/Await

Outcome

- ✓ Write clean, modern JavaScript required for React development.
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Module 2 – React JS Core Concepts

Understand how React builds modern Single Page Applications.

Topics Covered

- Introduction to React
- SPA Architecture
- JSX
- Components
- Functional Components
- Props
- State
- Event Handling
- Conditional Rendering
- Lists & Keys

Outcome

- ✓ Build reusable and interactive React components.
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Module 3 – React Hooks

Learn React's most powerful feature.

Topics Covered

- useState
- useEffect
- useRef
- useMemo
- useCallback

- useContext
- useReducer
- useEffect
- Custom Hooks
- useFetch
- useHttp
- useToggle
- useLocalStorage

Outcome

- ✓ Manage state, side effects, performance optimization, and reusable logic.
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Module 4 – React Router

Create professional multi-page applications.

Topics Covered

- BrowserRouter
- Route Configuration
- Navigation using Link
- Dynamic Routes
- Route Parameters
- Programmatic Navigation

Outcome

- ✓ Build complete navigation systems for React applications.
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Module 5 – Authentication & Authorization

Implement secure user authentication.

Topics Covered

- Login
- Registration

- JWT Authentication
- Protected Routes
- Role-Based Authorization
- Admin & User Access Control

Outcome

- ✓ Secure React applications using authentication and authorization.
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Module 6 – Backend Development with Express JS & PostgreSQL

Build REST APIs for React applications.

Topics Covered

- Express Server Setup
- PostgreSQL Database
- REST API Development
- CRUD Operations
- Database Integration
- Axios Communication

Outcome

- ✓ Connect React applications with a backend API and PostgreSQL database.
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Module 7 – Real-World JWT Authentication

Implement production-level authentication flow.

Topics Covered

- JWT Access Tokens
- Refresh Tokens
- Token Expiration
- Axios Interceptors

- Automatic Token Refresh
- Session Management
- Protected APIs

Outcome

- ✓ Understand and implement secure authentication used in enterprise applications.
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Module 8 – State Management using Redux Toolkit

Manage application-wide state efficiently.

Topics Covered

- Redux Store
- Slices
- Reducers
- createAsyncThunk
- Async API Calls
- Global State Management

Outcome

- ✓ Build scalable React applications using Redux Toolkit.
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Module 9 – API Integration & UI Development

Build professional-looking applications.

Topics Covered

- Axios Configuration
- Environment Variables
- API Integration

- Tailwind CSS
- Responsive Design
- Loading Indicators
- Toast Notifications
- SweetAlert2
- Error Handling

Outcome

- ✓ Create responsive and user-friendly React applications.
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Module 10 – Capstone Project



E-Commerce Application

Develop a complete real-world project from scratch.

Features

- User Registration
- Login System
- JWT Authentication
- Product Management
- Shopping Cart
- Razorpay Payment Integration
- Orders Management
- Admin Panel
- Protected APIs
- Responsive UI
- PostgreSQL Database
- Express REST APIs

Outcome

- ✓ Gain practical experience by building a complete production-ready Full Stack application.
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Technologies Covered

- HTML5
 - CSS3
 - JavaScript (ES6+)
 - React JS
 - React Router
 - React Hooks
 - Redux Toolkit
 - Axios
 - Tailwind CSS
 - Express JS
 - Node.js
 - PostgreSQL
 - JWT Authentication
 - Cookie-Based Authentication
 - REST APIs
 - Razorpay Payment Gateway
 - Git & GitHub (Optional)
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Learning Outcomes

After completing this course, participants will be able to:

- Develop modern React applications from scratch.
 - Build reusable and scalable React components.
 - Consume REST APIs using Axios.
 - Develop backend services with Express.js.
 - Integrate PostgreSQL databases.
 - Implement secure JWT authentication.
 - Manage global application state using Redux Toolkit.
 - Build responsive UIs with Tailwind CSS.
 - Integrate online payment gateways.
 - Develop and deploy a complete Full Stack E-Commerce application.
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Who Should Attend?

- Students
- Fresh Graduates
- Software Engineers

- Frontend Developers
 - Full Stack Aspirants
 - Professionals transitioning to React
 - Anyone interested in modern web development
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Prerequisites

- Basic Computer Knowledge
 - Basic HTML & CSS
 - Basic JavaScript Knowledge (preferred but not mandatory)
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Course Deliverables

- ✓ Complete Source Code
- ✓ Hands-on Coding Sessions
- ✓ Real-Time Project Development
- ✓ Interview-Oriented Explanations
- ✓ Industry Best Practices
- ✓ Course Completion Certificate (*Optional*)
- ✓ Lifetime Project Reference